NAME: ____________________________________________ CLASS: _________ INDEX: __________

# CATHOLIC JUNIOR COLLEGE
## JC2 PRELIMINARY EXAMINATION
## Higher 2

# BIOLOGY
**Paper 2 Structured Questions**

**9744/02**
**24 August 2021**

**2 hours**

Candidates answer on the Question Paper.

### READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| 1 [15] | |
| 2 [7] | |
| 3 [10] | |
| 4 [10] | |
| 5 [13] | |
| 6 [9] | |
| 7 [7] | |
| 8 [9] | |
| 9 [10] | |
| 10 [10] | |
| TOTAL | 100 |

Answer **all** questions.

**1** Fig. 1.1 shows an electron micrograph of a cell.

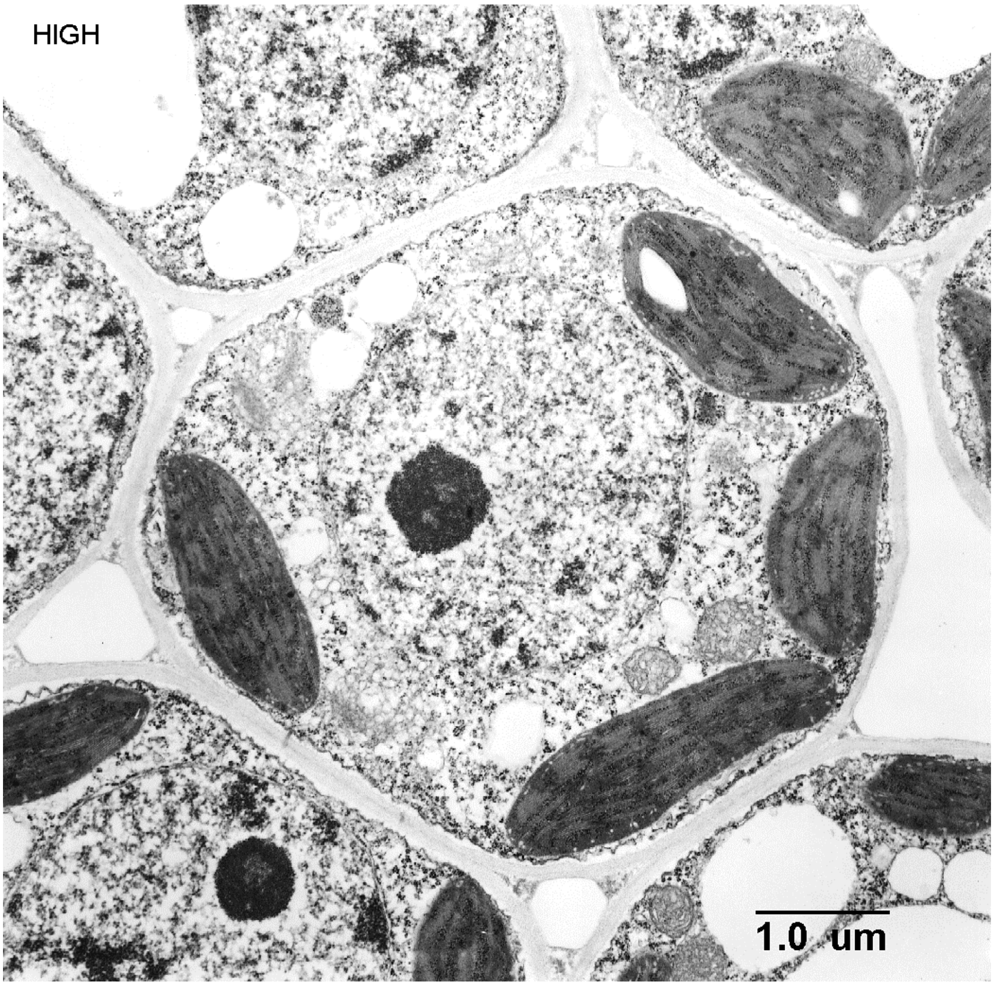


**Fig. 1.1**

**(a)** **(i)** With reference to the Fig 1.1, state two reasons why the cell is considered to be a eukaryotic cell.

Use 2 separate label lines, labelled **1** and **2**, to clearly mark out features visible in Fig. 1.1 to support your answer.

**1** ……………………………………………………………………………………………….…………

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**2** ………………………………………………………………………………………………………..

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**(b)** Fig. 1.2 shows the structure of an adenovirus, which causes a wide range of illnesses such as common cold or flu-like symptoms, fever and sore throat.

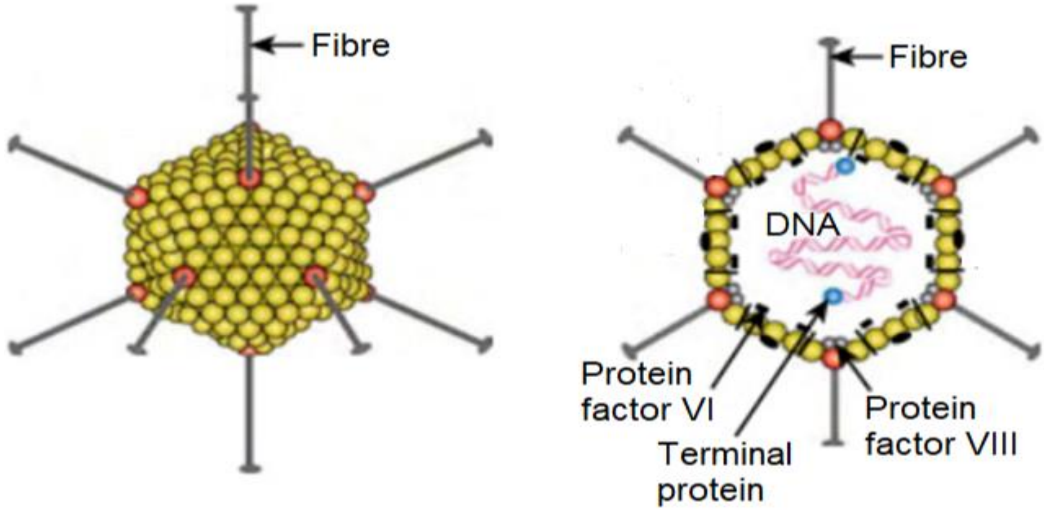


**Fig. 1.2**

**(i)** Outline the cell theory.

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**(ii)** With reference to Fig. 1.2, explain how this virus challenges the cell theory.

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Stachyose is a storage oligosaccharide found in beans, peas and other legumes.

Stachyose can only be metabolized by anaerobic microorganisms in the large intestine. It is responsible for the production of carbon dioxide, hydrogen sulfide and other gases which collectively make up what is known as flatulence.

The structure of stachyose is shown in Fig. 1.3.

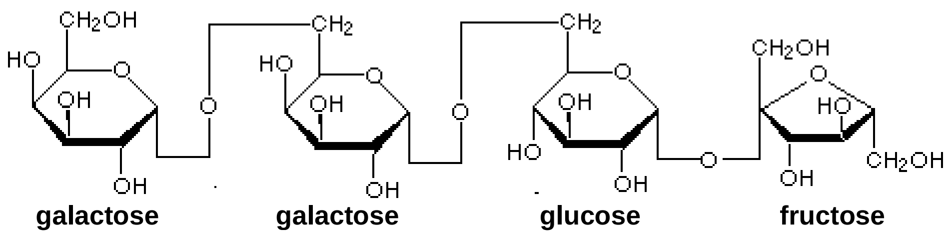


**Fig. 1.3**

**(c)** Define the term *glycosidic bond*.

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**(d)** Several commercial products are available to assist in the digestion of oligosaccharides like stachyose.

Suggest how such products prevent the production of flatulence.

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**(e)** When small amounts of paper containing cellulose are ingested by children, there is usually no gas production.

Suggest one reason and a possible consequence of such an event.

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An enzyme known as α-galactosidase cleaves α-(1,6)-glycosidic bonds to sequentially release terminal galactose residues.

The breakdown of stachyose by α-galactosidase is a two-step process. In the first step of this process, stachyose is hydrolysed to galactose and raffinose.

**(f)** With reference to Fig. 1.3, draw the products of the first step in this hydrolysis.

[2]

Fig. 1.4 illustrates the activity of α-galactosidase on the substrate stachyose over a period of time.

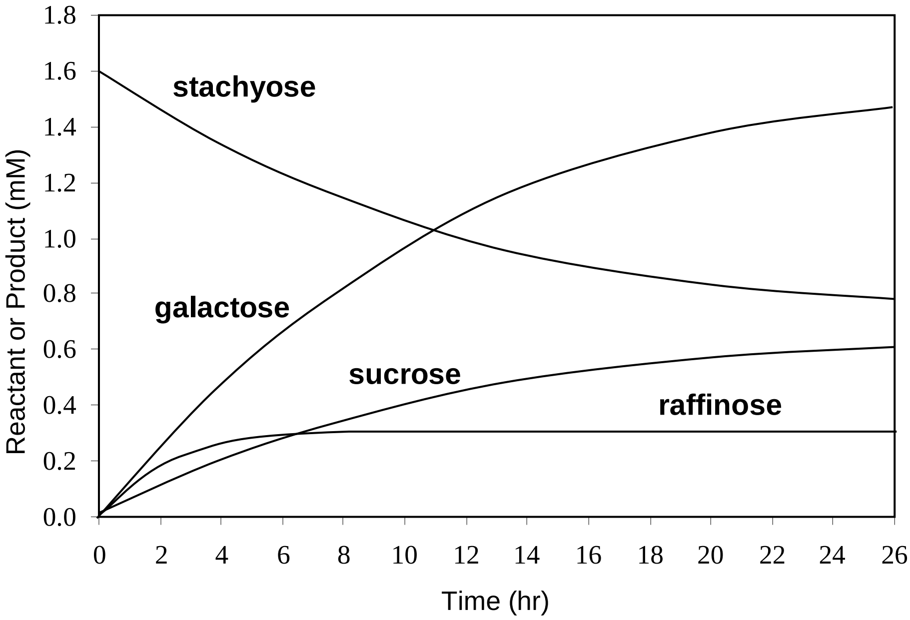


**Fig. 1.4**

**(g)** With reference to Fig. 1.4, explain why the level of raffinose remains relatively constant.

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**(h)** Suggest why the rate at which stachyose is broken down decreases as the concentration of galactose increases.

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[Total: 15]

**2** **(a)** The “trombone” model of DNA replication postulates that two DNA polymerase enzymes work together in a protein complex.

In Fig. 2.1, structures labelled **P, Q** and **R** are enzymes involved in the process of DNA replication. Connecting proteins join the two DNA polymerase molecules to enzyme **P**, forming the protein complex.

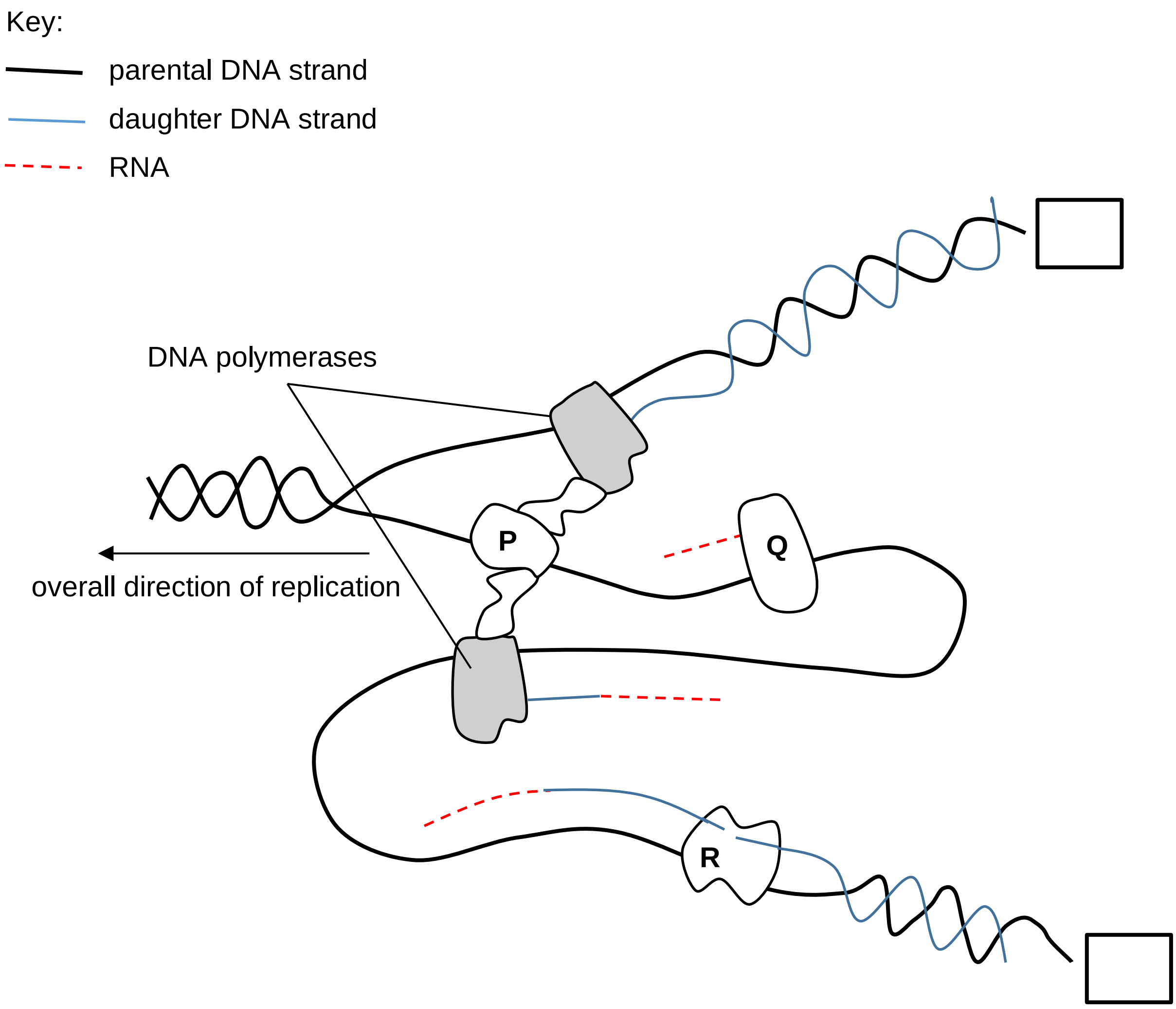


**Fig. 2.1**

**(i)** Complete the boxes to indicate clearly the 5’ and 3’ ends of the parental strands shown in Fig. 2.1. [1]

**(ii)** Identify enzymes **P, Q** and **R**. [1]

**P**: ......................................................................

**Q**: ......................................................................

**R**: ......................................................................

Fig 2.2 illustrates the binding of a protein to a certain region of a DNA molecule

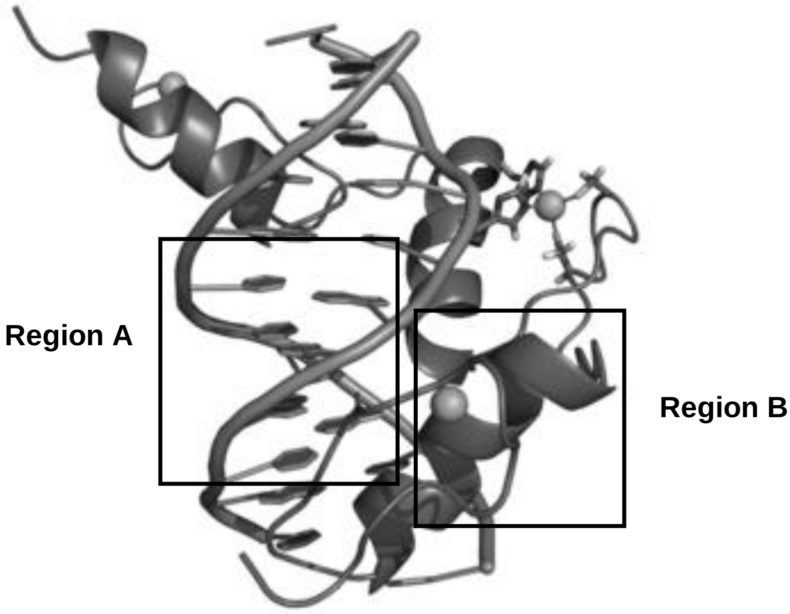


**Fig 2.2**

**(b)** With reference to Fig. 2.2, describe how the helical structure shown in **Region A** differs from that of **Region B**.

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Fig. 2.3 is a diagram of a telomere showing a unique single stranded G-rich 3' overhang that loops back to form a hairpin structure.

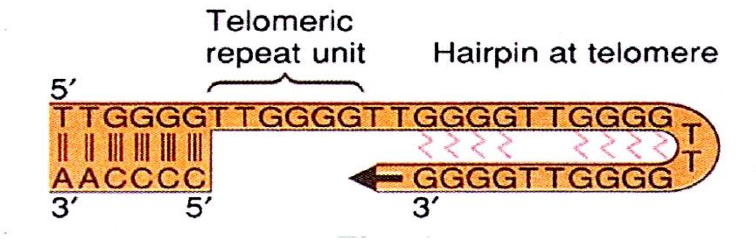


**Fig.2.3**

**(c)** Outline the role(s) of telomeres in DNA replication.

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[Total: 7]

**3** Fig. 3.1 shows the structure of molecules and some associated processes found in a eukaryotic cell.

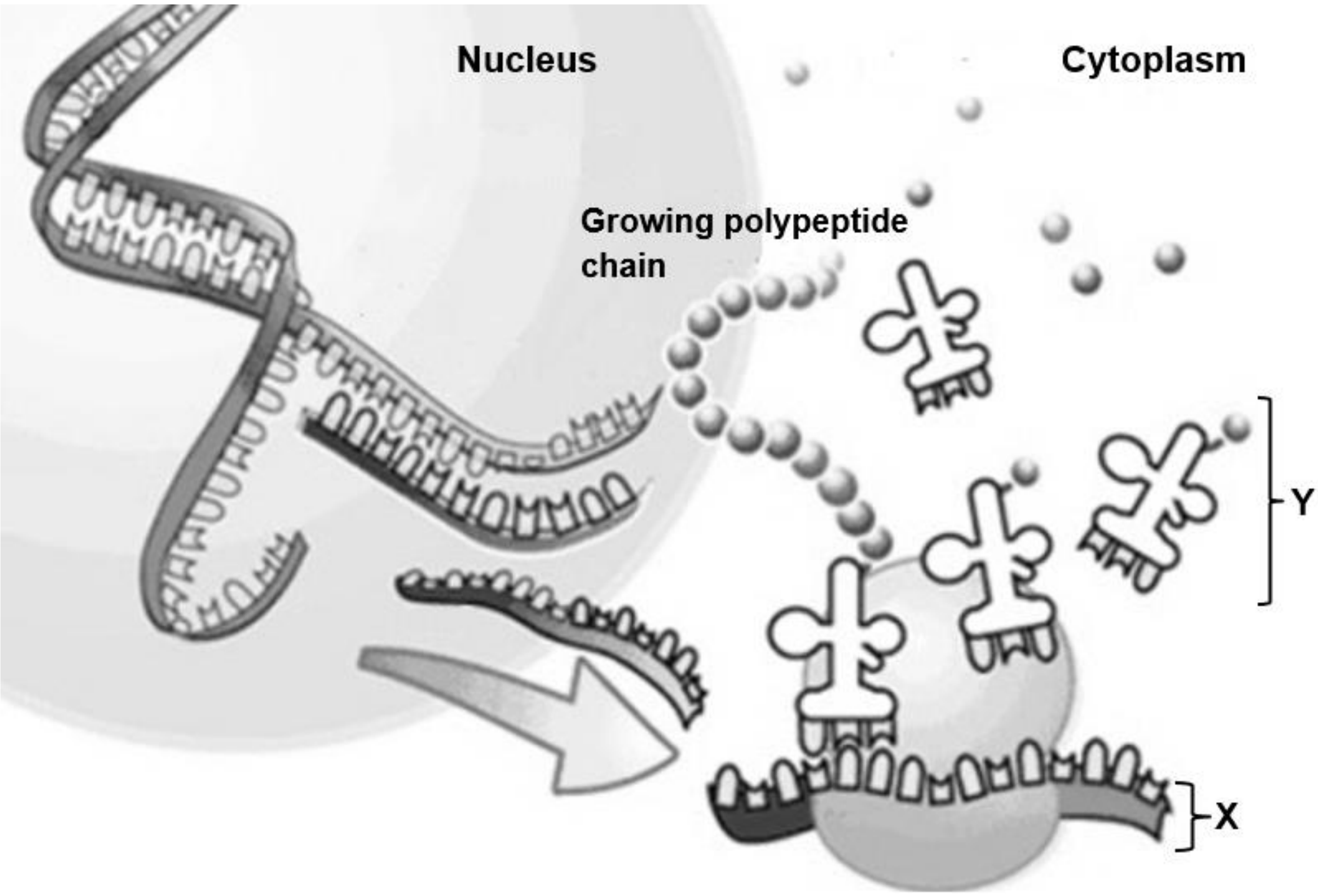


**Fig. 3.1**

**(a)** With reference to Fig. 3.1, outline the events that occurred in the nucleus that synthesised structure **X**.

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**(b)** Identify structure **Y**.

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**(c)** Parasites cause diverse types of disease, for example, malaria.

Drug treatments are required to address varying causes of pathogenesis. A new class of drugs has been deployed to block the formation of structure **Y** in parasites.

Suggest how the use of such drugs is effective in treating parasites infections.

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There is an absent of nucleus in bacteria. In addition, genes that are involved in the same metabolic pathway are usually arranged together in a structure called operon.

An example of an operon is the *lac* operon.

**(d)** Explain how the location of the *lac* operator allow it to regulate the expression of structural genes in *lac* operon.

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**(e)** Explain how the presence of high concentration of glucose results in low expression of structural genes found in *lac* operon despite the absence of any active *lac* repressor proteins.

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**(f)** Some bacteria are found to have a loss of function mutation in the *lac* operator.

When grown in an environment containing high concentration of glucose and no lactose, these bacteria grow slower than the bacteria carrying normal functioning *lac* operator. Suggest why.

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[Total: 10]

**4** Epidermal growth factor (EGF) is a protein composed of 53 amino acids. It binds to specific cell surface membrane receptors to promote cell proliferation.

**(a)** Based on the information above**,** explain why EGF cannot move across cell surface membrane through channel proteins.

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HER2 receptors are glycoproteins found on cell surface membrane. They are a type of tyrosine kinase receptor that recognises EGF.

**(b)** Explain how the structure of HER2 receptor allows it to be embedded across cell surface membrane and serve as a transmembrane protein.

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**(c)** Name another ligand molecule that binds to tyrosine kinase receptors and regulate blood glucose level.

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Fig. 4.1 below briefly illustrates HER2 signalling pathway.

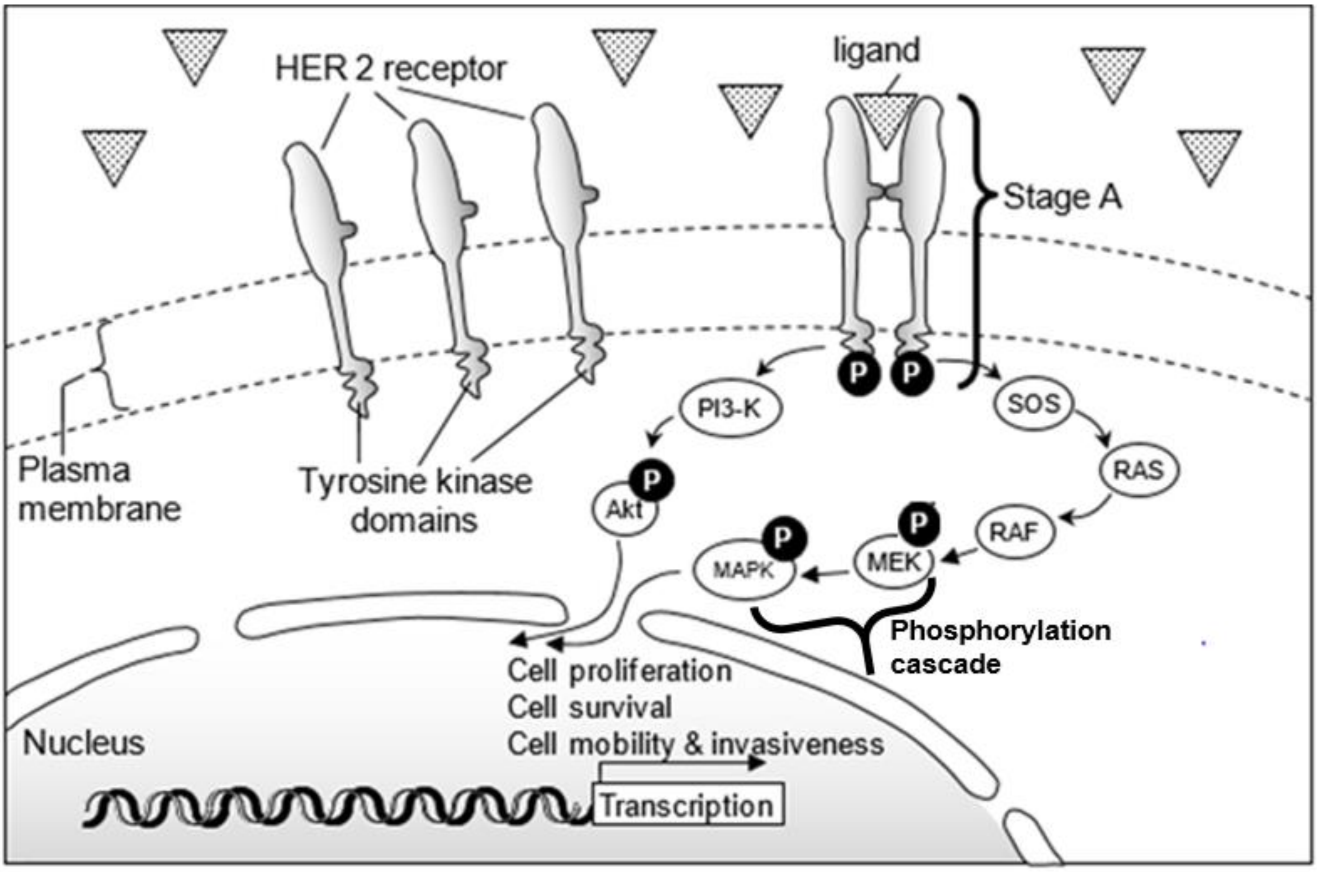


**Fig. 4.1**

**(d)** Explain the events occurring in stage A of the HER2 signalling pathway shown in Fig. 4.1.

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**(e)** RAF (an example of protein kinase) can be activated by phosphorylation. It can be activated by direct interaction with its allosteric site too.

Fig. 4.1 shows that RAS protein activates RAF by direct interaction instead of phosphorylation.

Suggest why RAS does **not** activate RAF by phosphorylation.

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[Total: 10]

**5** Fig. 5.1 shows the role of stem cells in the immune system and the variety of cell types they are able to differentiate into.

Fig 5.2 shows a functional group found in the nucleus of eukaryotic cells.

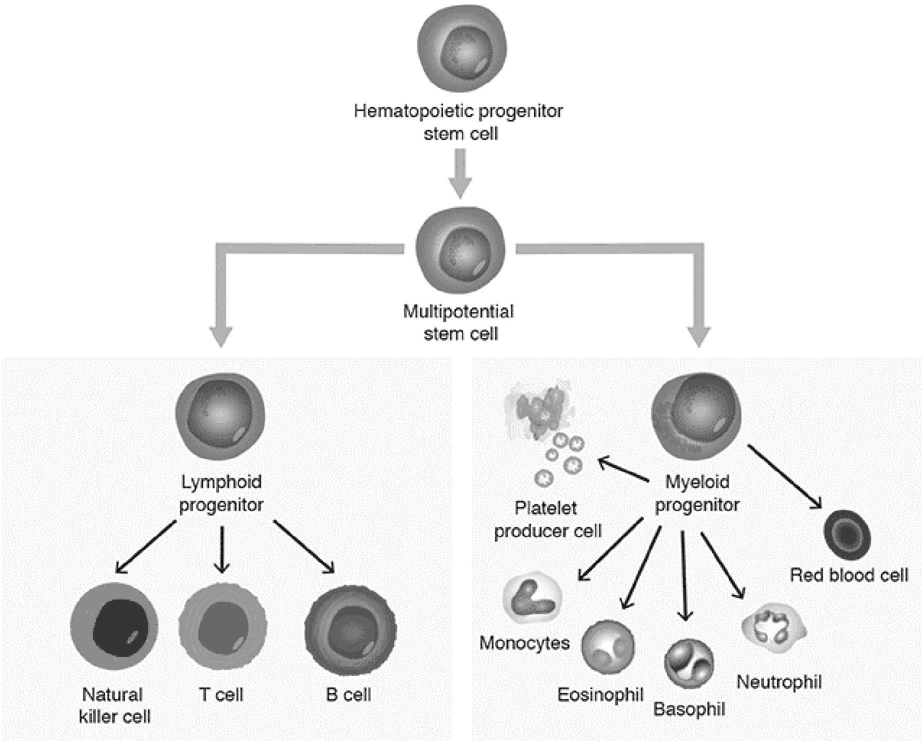


**Fig. 5.1**

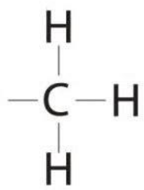

**Fig. 5.2**

**(a)** Name one source of totipotent stem cells.

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**(b)** Explain the role of the functional group shown in Fig. 5.2 in determining the differentiation of stem cells into specific cell types.

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**(c)** With reference to the control of cell cycle, explain how the initiation of cell division is controlled in stem cells.

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**(d)** In an active organ such as liver, the rate of transcription in the cells is usually high.

With reference to the control of gene expression at transcriptional level, explain fully how the high rate of transcription in these cells can be achieved.

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**(e)** Suggest why specific transcription factors are considered an efficient way of control of gene expression in eukaryotic genes.

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[Total: 13]

**6** In human, hereditary breast and ovarian cancer syndrome (HBOC) is an inherited genetic condition that follows an autosomal dominant inheritance.

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HBOC is associated with loss of function mutations in *BRCA* genes which are a group of tumour suppressor genes. The products of these genes are involved in repairing of damaged DNA and initiating apoptosis in cells where the DNA damage is beyond repair.

Fig. 6.1 below shows the percentage of occurrence of breast cancer and ovarian cancer in women who are *BRCA* mutation carriers (women with HBOC) and women in the general population (women without HBOC).

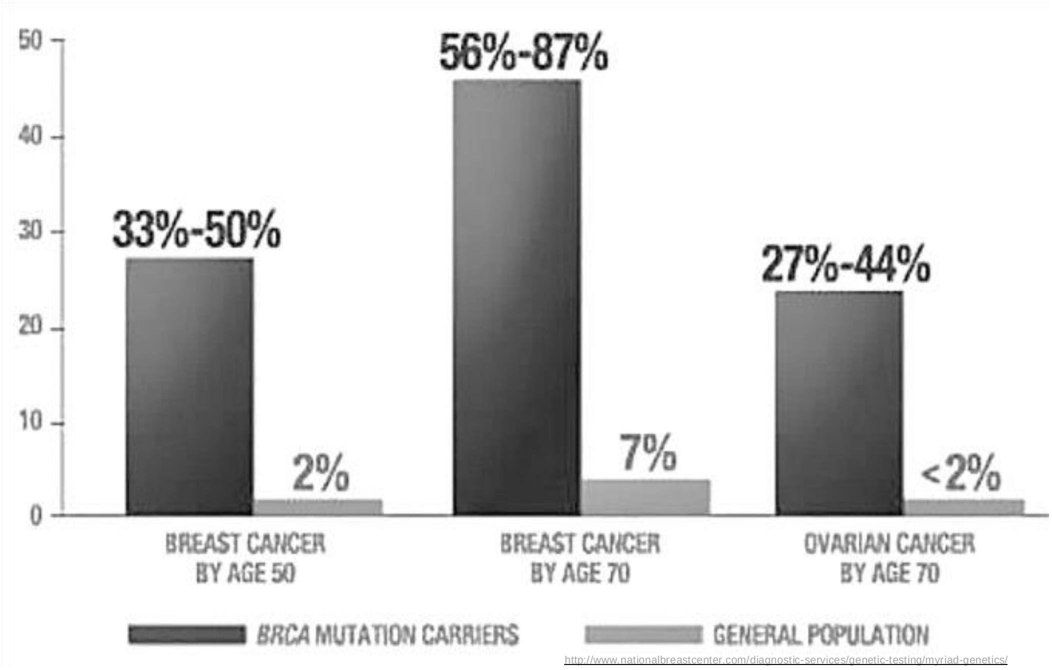


**Fig. 6.1**

**(a)** With reference to Fig. 6.1, explain the difference in percentage occurrence of breast cancer between women who are *BRCA* mutation carriers and women in the general population by the age of 50.

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**(b)** With reference to the women in general population in Fig. 6.1, explain the difference in percentage occurrence of breast cancer between women by the age of 50 and by the age of 70.

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**(c)** With reference to Fig. 6.1,

**(i)** describe the difference in percentage occurrence of ovarian cancer and percentage occurrence of breast cancer in women who are *BRCA* mutation carriers.

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………………………………………………………………………………………………….. [1]

**(ii)** suggest a possible reason for the difference described in part **c(i)**

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**(d)** Scientists concluded that women who are *BRCA* mutation carriers (women with HBOC) may not necessarily get ovarian cancer.

Use the information in Fig.6.1 to justify this conclusion.

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[Total: 9]

**7** The figure below shows eukaryotic chromosomes.

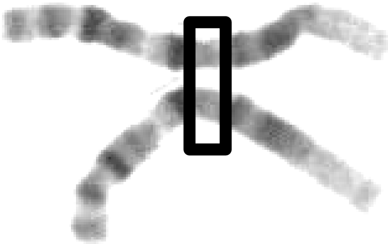

**Fig 7.1**

**(a)** Name the region shown and explain its role in full.

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The Canine Genome Sequencing Project was successfully completed in year 2005. During the project, researchers needed to investigate two particular genes in dogs.

The genes were amino-methyl-transferase (*AMT*) gene and T-cell leukemia translocation-associated (*TCTA*) gene.

*AMT* gene codes for AMT enzyme that degrades excess glycine.

*TCTA* gene codes for TCTA protein involved in the formation of osteoclasts (osteoclastogenesis) for the breakdown of bone tissue.

**(b)** To increase sample size for statistical reliability, scientists selected several heterozygous female poodles for normal AMT phenotype and normal TCTA phenotype, and conducted test crosses on them. The following results were observed in the offspring.

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|---|---|
| Normal AMT, normal TCTA dogs | 96 |
| Normal AMT, mutant TCTA dogs | 18 |
| Mutant AMT, normal TCTA dogs | 20 |
| Mutant AMT, mutant TCTA dogs | 94 |

**(i)** Using **A** and **<u>a</u>** to represent the alleles for the *AMT* gene, and **T** and **<u>t</u>** to represent the alleles for the *TCTA* gene, illustrate the above cross with the use of a genetic diagram.

[4]

**(ii)** Explain the low frequency of recombinants observed in the offspring.

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…………………………………………………………………………………………..…….. [1]

[Total: 7]

**8** Human T- lymphotropic virus type 1 (HTLV-1) is a retrovirus that brings about a form of cancer called T-cell leukaemia.

HTLV-1 has a similar reproductive cycle as human immunodeficiency virus (HIV) and predominantly infects CD4$^+$ T cells too.

Fig. 8.1 shows the structure of a mature HTLV-1 particle.

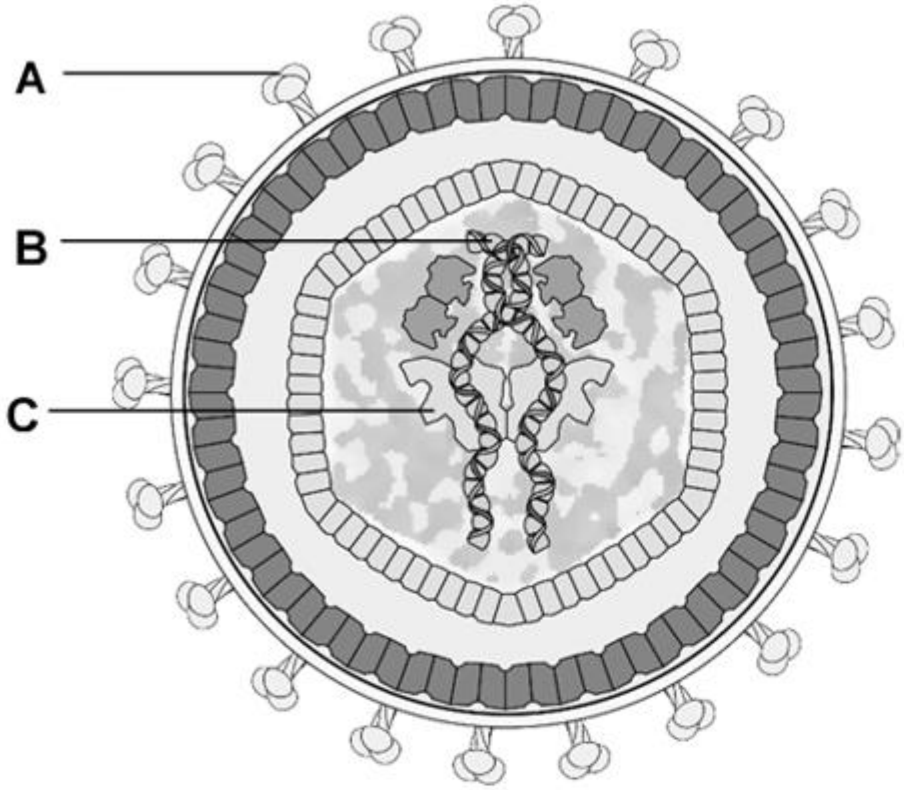


**Fig. 8.1**

**(a)** With reference to Fig. 8.1;

**(i)** Name structure **C**, which is the first viral enzyme involved in the reproductive cycle of HTLV-1

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**(ii)** Describe the roles of structures **A** and **B** in the reproductive cycle of HTLV-1 in CD4$^+$ T cells.

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**(b)** Suggest how HTLV-1 infection could lead to the onset of T-cell leukaemia.

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**(c)** Explain why antibiotics are not used to treat HTLV-1 infection.

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**(d)** Describe how HTLV-1 infection induces the production of antibodies in the body.

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**(e)** During the secondary immune response, majority of the antibodies secreted and circulating in the body is IgM.

Lumens of organs are mainly lined with epithelium. During secondary infection, majority of antibodies found within the lumen of organs are IgA instead of IgM. Suggest why.

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[Total: 9]

**9** The polar bear (*Ursus maritimus*) is most closely related to the brown bear (*Ursus arctos*). The brown bear is widely distributed across Holoarctic habitats (land masses surrounding but not part of the Arctic Circle). The two species differ in morphology, ecology and behaviour, reflecting adaptations to different ecological niches.

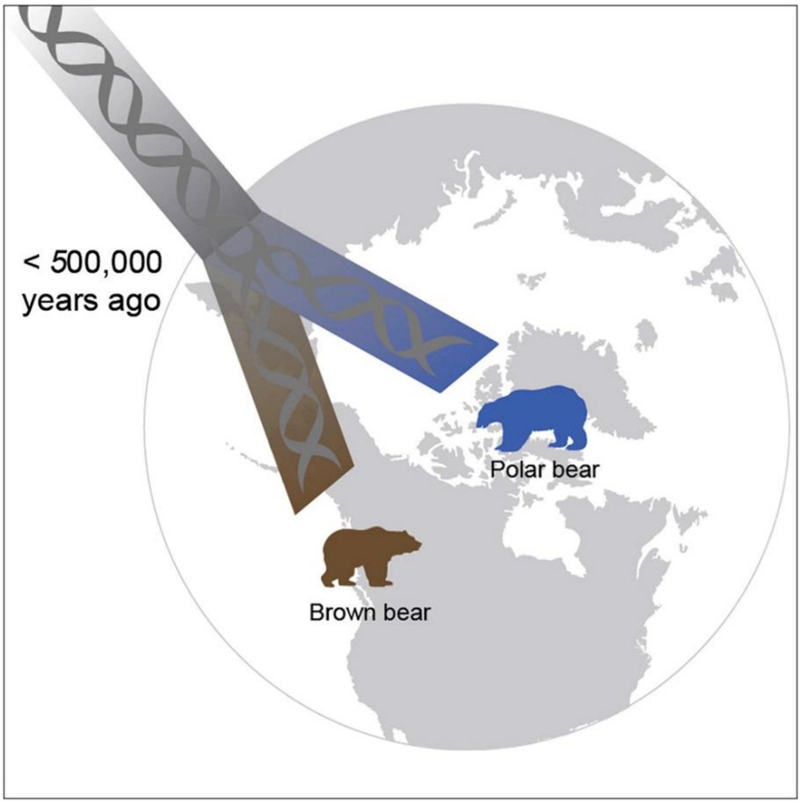


**Fig. 9.1**

The land masses that make up the polar bears habitat were once connected to North America, but over the past 60 million years these land masses have slowly shifted up in latitude till the Arctic Circle.

**(a)** Explain how polar bears diverged into a separate species from brown bears.

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Fossil of the bears ancestors' found in the Arctic caused scientists to initially estimated the divergence of these two species at 800 000 years ago. Recent mitochondrial genome sequencing data has suggested that the two species diverged less than 500 000 years ago, with polar bears exclusively colonising the Arctic Circle.

**(b)** Explain one advantage of using nucleotide sequences over morphology to establish phylogenetic relationships.

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During the winter months bears hibernate as conditions in the environment become too extreme. Studies have shown that during hibernation, bear metabolism shows a preference for lipids as the energy source instead of carbohydrates.

**(c)** When triglycerides are hydrolysed, glycerol can be converted to a product that is an intermediate in glycolysis.

Suggest what glycerol is converted into such that it can enter the glycolysis pathway.

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**(d)** At certain times of the day oxygen levels dip, causing the hibernating bears to switch to anaerobic respiration for a short while.

Explain the significance of conversion of pyruvate into lactate during anaerobic respiration.

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[Total: 10]

**10** Coral reefs are a vital source of income for fishermen. They provide fish fry with shelter and abundant food. They also protect coastlines from storms and being eroded. Yet each year, the range of coral cover is decreasing and its reason has been attributed to the enhanced greenhouse effect and climate change.
Fig.10.1 below shows the trend of sea surface temperature over the Great Barrier Reef from 1900 till 2020.

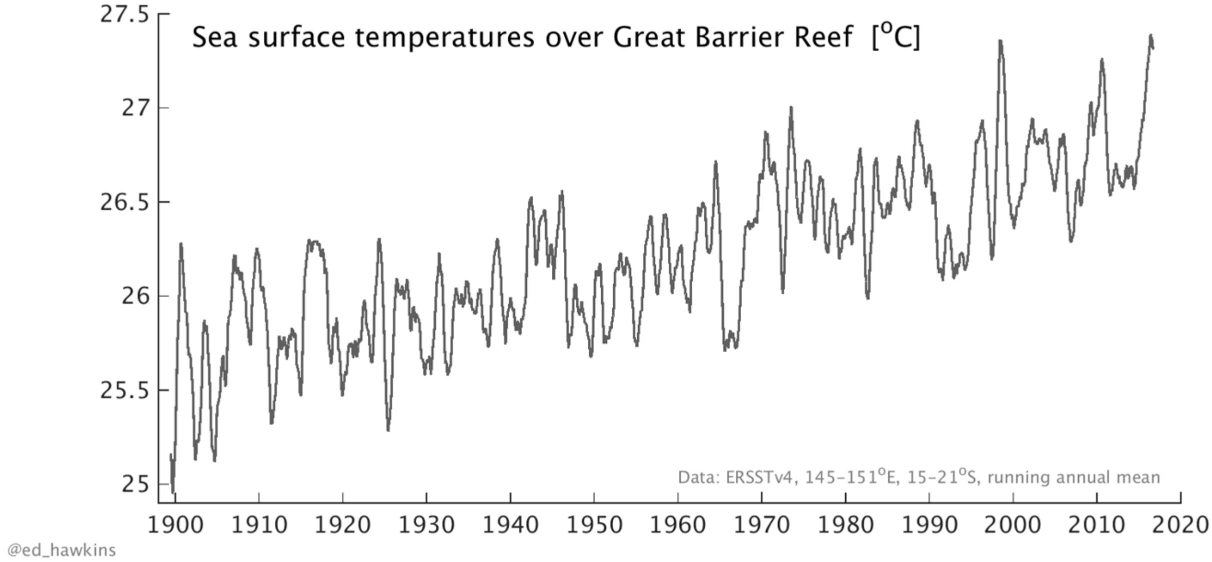


**Fig.10.1**

**(a)** Describe a human activity that has contributed to the enhanced greenhouse effect.

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**(b)** With reference to Fig.10.1, explain how the rise in ocean temperatures is linked to the reduction in coral reef cover.

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Plants found in the sea like kelp, seaweed and seagrass depend on photosynthesis for food production.

In summer months where the days are longer, there is an increase in carbon dioxide concentration of oceans. Population of sea plants are expected to rise.

**(c)** Explain why the increase in sea plants is expected in the summer months.

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**(d)** Herbicides runoff from coastlines can cause sea plants to die and decrease in population. Many herbicides are photosystem inhibitors.

Explain briefly how herbicides cause the death of sea plants.

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[Total: 10]

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